Small-energy regularity for stationary harmonic maps

A complete proof for compact embedded C^2 targets

Abstract

Let N be a nonempty compact embedded C^2 submanifold without boundary in Euclidean space. In every domain dimension $n \geq 2$, a stationary weakly harmonic map whose normalized energy is sufficiently small on a ball has a unique continuous N-valued representative on the concentric half ball. We prove the scale-correct estimate with Hölder exponent $1/2$ and the square root of the original normalized energy. The proof uses the normal projection antisymmetric system, a natural Coulomb frame with critical Morrey control, compensation for the actual bounded cutoff of the compact-target map, and an all-center decay iteration. The frame is constructed by stronger Morrey perturbations, boundary absorption, and a joint critical limit. The Hodge fields are selected once and retained through the estimates. No minimizing property and no separately assumed gauge, decay estimate, or regular representative enter the theorem. For a certified ambient normal-projection extension with rational derivative bound ℓ, explicit rational functions of n, q, ℓ provide an admissible energy threshold and Hölder coefficient. A complete arithmetic ledger and executable exact-arithmetic algorithm specify these conservative constants; a final correspondence table identifies their formal proofs.

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1 The exact statement and weak conventions

Fix integers $n \geq 2$, $q \geq 0$, and a nonempty compact embedded C^2 submanifold $N \subset \mathbb{R}^q$ without boundary. No orientation, global normal frame, or smoothness beyond C^2 is required. Gradient energies and matrix-valued one-form energies use Hilbert–Schmidt norms. Norms of linear and bilinear maps are induced operator norms. In particular, $\|DG(y)\|$ is the operator norm from Euclidean $\mathbb{R}^q$ to the operator-norm space $\mathcal{L}(\mathbb{R}^q, \mathbb{R}^q)$. A weak gradient is the distributional gradient of the original map, and is kept fixed throughout. Set

$$E(u; V) = \int_V |Du|^2 \, \mathrm{d}x, \qquad \Theta_u(a, r) = r^{2-n} E(u; B_r(a)).$$

There is no factor $1/2$ in the energy. For $y \in N$, let $Q(y)$ be the orthogonal projection onto $(T_y N)^\perp$ and $\Pi(y) = I - Q(y)$. The map Q is C^1 along N and admits a compactly supported C^1 ambient extension $\widehat{Q}$ with bounded derivative. Fix any such extension $G = \widehat{Q}$ and a rational $\ell \geq 0$ such that $\sup_{y \in \mathbb{R}^q} \|DG(y)\| \leq \ell$. Compactness guarantees the existence of this pair; numerical evaluation below uses a certified pair. Use the sign convention $\mathcal{A}_y(v, w) = DQ(y)[v]w$ for tangent vectors v, w.

Definition 1.1. A map $u \in W^{1,2}_{\mathrm{loc}}(U, \mathbb{R}^q)$, with $u \in N$ almost everywhere, is weakly harmonic if the following holds for every $V \in C^1_c(\mathbb{R}^n \times \mathbb{R}^q, \mathbb{R}^q)$ whose support is contained in $U \times \mathbb{R}^q$ and for which $V(x, y) \in T_y N$ for every $x \in \mathbb{R}^n$, $y \in N$: the integrand below is integrable and

$$\int_U \sum_i \langle \partial_i u(x), DV(x, u(x))[e_i, \partial_i u(x)] \rangle \, \mathrm{d}x = 0. \tag{1.1}$$

This is the complete ambient map-variation condition. In particular, after a compact ambient cutoff equal to one near N, it includes the variation $V(x, y) = \Pi(y)\varphi(x)$ for every compactly supported ambient C^1 field φ. The resulting weak equation is derived in Lemma 2.1. The map is domain stationary if, in addition,

$$\int_U \left(|Du|^2 \operatorname{div} X - 2 \sum_{i,j=1}^{n} \langle \partial_i u, \partial_j u \rangle \, \partial_i X_j \right) \mathrm{d}x = 0 \quad (X \in C^\infty_c(U, \mathbb{R}^n)). \tag{1.2}$$

Theorem 1.2 (Small-energy regularity)**.** *Every fixed embedded target $N \subset \mathbb{R}^q$ above admits constants $\varepsilon_* > 0$ and $C < \infty$, depending only on n, N, with the conclusion below. More precisely, given any certified compactly supported C^1 extension $G : \mathbb{R}^q \to \mathcal{L}(\mathbb{R}^q, \mathbb{R}^q)$ and rational $\ell \geq 0$ satisfying*

$$G(y) = Q(y) \quad (y \in N), \qquad \|DG(y)\| \leq \ell \quad (y \in \mathbb{R}^q),$$

take the positive rational gauge constants κ, C_g defined exactly in (7.6) and (7.5), and the positive rational given-frame constants θ, C_r defined exactly in (7.7). Their complete arithmetic

definitions are in Appendix A; all four depend only on n, q, and $\theta \le 1$. Explicit admissible constants and exponent are

$$M = (1 + 4q\ell^2)4^{n-2}, \qquad K = (1 + C_g)(1 + M),$$
$$\varepsilon_* = \left(\frac{\min(\kappa, \theta)}{K}\right)^2, \qquad C = 2C_r K, \qquad \alpha = \frac{1}{2}. \tag{1.3}$$

These constants are fully computable from n, q, ℓ after the geometric certificate has been supplied. Existence of a certificate for each abstract N gives the original target-only existence statement; numerical evaluation uses certified geometric data as input.

Suppose $U \subset \mathbb{R}^n$ is open, $u \in W^{1,2}_{\mathrm{loc}}(U, \mathbb{R}^q)$ takes values in N almost everywhere, and u is weakly harmonic and domain stationary in the sense of Definition 1.1. If $R > 0$, $\overline{B_R(a)} \subset U$, and

$$\Theta_u(a, R) \le \varepsilon_*, \tag{1.4}$$

there is a continuous representative $\widetilde{u} : B_{R/2}(a) \longrightarrow N$ satisfying

$$|\widetilde{u}(x) - \widetilde{u}(y)| \le CR^{-1/2}\Theta_u(a, R)^{1/2}|x - y|^{1/2} \qquad (x, y \in B_{R/2}(a)). \tag{1.5}$$

It is unique among all continuous $\mathbb{R}^q$-valued representatives of u on the half ball. The constants precede the domain, map, center, and radius.

The argument follows the antisymmetric-potential and Morrey-frame method of Rivière and Struwe [1, 2]. Its analytic application here is to the compact-target map itself. We do not assert the wider regularity theorem for arbitrary, possibly unbounded, Euclidean-valued solutions of antisymmetric systems. Compactness supplies a genuine bound for a cutoff used in the native compensation pairings; the estimates do not depend on the size of this bound.

The accompanying Lean theorem has precisely the stated hypotheses, with the original explicit weak gradient. Its proof has no unformalized analytic assumption. The prose uses ordinary background real analysis (Lebesgue differentiation, weak Sobolev calculus, Hilbert projection, and the Banach inverse function theorem) and gives the analytic constructions needed here. Section 10 identifies their formal counterparts.

2 Geometry and stationarity

Lemma 2.1 (Tangency and the original weak equation)**.** *One has $Q(u)\partial_i u = 0$ almost everywhere and*

$$\int_U Du : D\varphi = \int_U \sum_i \mathcal{A}_u(\partial_i u, \partial_i u) \cdot \varphi \quad (\varphi \in C^\infty_c(U, \mathbb{R}^q)). \tag{2.1}$$

The skew one-form

$$\Omega_i = Q(u)\partial_i(Q \circ u) - \partial_i(Q \circ u)Q(u) \tag{2.2}$$

satisfies

$$-\Delta u = \sum_i \Omega_i \partial_i u, \qquad |\Omega| \le 2\sqrt{q}\,\ell|Du|. \tag{2.3}$$

Proof. Local defining functions for the embedded manifold and the C^1 Sobolev chain rule show that Du is tangent almost everywhere. The same rule gives $D(\widehat{Q} \circ u) = D\widehat{Q}(u)Du$. Use the

tangent test $\Pi(u)\varphi$ in the first variation. Because $\Pi(u)Du = Du$, $D\Pi = -DQ$, and DQ is bounded on the compact target, the product rule gives

$$0 = \int_U Du : D(\Pi(u)\varphi) = \int_U Du : D\varphi - \int_U \sum_i DQ(u)[\partial_i u]\partial_i u \cdot \varphi.$$

These bounded native tests are legitimate: bounded smooth approximations of their weak graphs converge strongly in $W^{1,2}$, with a uniform value bound obtained by Lipschitz truncation. The derivative terms converge by Holder, and the L^1 forcing terms by dominated convergence. This proves the equation from the original variations.

Both $Q(u)$ and its derivatives are symmetric; their commutator is skew. Differentiating $Q^2 = Q$ shows that $DQ(y)[v]w$ is normal for tangent v, w. Therefore $\Omega_i \partial_i u = Q(u)\partial_i(Q \circ u)\partial_i u = \partial_i(Q \circ u)\partial_i u$. The bound for $D\widehat{Q}$ and the preceding weak equation complete the proof, with the sign displayed in (2.3). □

Lemma 2.2 (Exact monotonicity)**.** *If $0 < r < s$ and $\overline{B_s(a)} \subset U$, then*

$$\Theta_u(a, s) - \Theta_u(a, r) = 2 \int_{B_s(a)\setminus B_r(a)} |x - a|^{2-n} |\partial_{\mathrm{rad},a} u|^2 \, \mathrm{d}x \geq 0, \tag{2.4}$$

where $\partial_{\mathrm{rad},a} u = Du(x)(x - a)/|x - a|$ away from a.

Proof. Translate a to zero and put $X(x) = \eta(|x|)x$ in (1.2), first with η smooth and constant near zero. This gives

$$\int \big((n-2)\eta|Du|^2 + |x|\eta'\big(|Du|^2 - 2|\partial_{\mathrm{rad}} u|^2\big)\big) = 0.$$

For $F(t) = \int_{B_t} |Du|^2$, polar integration and radial approximation yield, for almost every $t > 0$,

$$(n-2)F(t) = tF'(t) - 2t \int_{\partial B_t} |\partial_{\mathrm{rad}} u|^2 \, \mathrm{d}\mathcal{H}^{n-1}.$$

F is absolutely continuous on every compact positive radius interval. Differentiate $t^{2-n}F(t)$, integrate, and use polar coordinates again. This proves the equality for every positive pair of radii, also for $n = 2$. No minimizing comparison has been used. □

For a field on a ball D, use the complete truncated critical quantity

$$\mathcal{M}_2(F; D)^2 = \sup_{x \in D,\, t > 0} t^{2-n} \int_{B_t(x) \cap D} |F|^2 \, \mathrm{d}x. \tag{2.5}$$

Lemma 2.3 (All-center control from one energy)**.** *If $e = \Theta_u(a, R)$ and $D = B_{3R/4}(a)$, then*

$$\mathcal{M}_2(Du; D)^2 + \mathcal{M}_2(\Omega; D)^2 \leq Me, \qquad M = (1 + 4q\ell^2)4^{n-2}. \tag{2.6}$$

Proof. For $x \in D$ and $t \leq R/4$, monotonicity at x between t and $R/4$, and $B_{R/4}(x) \subset B_R(a)$, give $t^{2-n} \int_{B_t(x)\cap D} |Du|^2 \leq 4^{n-2}e$. For $t > R/4$, bound t^{2-n} by $(R/4)^{2-n}$ and use the full energy on $B_R(a)$. Finally $|\Omega| \leq C_N |Du|$. □

3 Ball projections and native compensation

An interior divergence-free L^2 field Z satisfies $\int Z : D\psi = 0$ for compactly supported tests. A *natural* divergence-free field satisfies this for every $\psi \in W^{1,2}(D)$, including its weak zero normal flux. The distinction matters below.

3.1 One Dirichlet and natural decomposition

Lemma 3.1 (Ball projections)**.** *Let $D = B_r(a)$ and $F \in L^2(D, \mathbb{R}^q \otimes \Lambda^1)$. There are $f \in W_0^{1,2}(D, \mathbb{R}^q)$ and a mean-zero $w \in W^{1,2}(D, \mathbb{R}^q)$ uniquely characterized by*

$$\int_D Df : D\varphi = \int_D F : D\varphi \qquad (\varphi \in W_0^{1,2}(D)), \tag{3.1}$$

$$\int_D Dw : D\psi = \int_D F : D\psi \qquad (\psi \in W^{1,2}(D)). \tag{3.2}$$

Set $Q = F - Dw$, $h = Dw - Df$. Then

$$F = Df + Q + h, \tag{3.3}$$

Q is natural divergence-free, h is closed and divergence-free with harmonic coefficients, and the three fields are mutually L^2 orthogonal. Each projection is bounded on every L^s, $1 < s < \infty$, with constants depending only on n, s. If $p' = p/(p-1)$, then

$$\|Df\|_{L^p(D)} \le C_{n,p} \sup_{\substack{\varphi \in W_0^{1,p'}(D) \\ \|D\varphi\|_{p'} \le 1}} \left| \int_D F : D\varphi \right|. \tag{3.4}$$

For $0 < \theta \le 1/2$ one also has

$$\int_{B_{\theta r}(a)} |F|^p \le C_{n,p} \left(\|Df\|_p^p + \|Q\|_p^p + \theta^n \|F\|_p^p \right). \tag{3.5}$$

Proof. Poincare makes the gradient norm coercive on the zero-boundary and mean-zero spaces. Hilbert projection gives (3.1) and (3.2). Subtraction for compactly supported tests gives $\operatorname{div} h = 0$, and $dh = 0$ since h is a difference of gradients. Thus its coefficients are harmonic. The defining identities give all three orthogonalities.

Here is the analytic mechanism for the all-power bounds. The gradient response kernels of the ball Dirichlet and mean-zero Neumann problems satisfy

$$|K(x,y)| \le C|x-y|^{-n}, \qquad |K(x,y) - K(x',y)| \le C|x-x'|\,|x-y|^{-n-1}$$

when $2|x-x'| \le |x-y|$. These follow from the Green response and harmonic gradient estimates, including the matching homogeneous boundary comparison on a spherical cap. One may obtain the latter by inversion in the sphere with the normal-flux transport for the natural problem. Split a source at twice the radius of a test ball. The near response is controlled by the L^2 projection norm; the far response is harmonic and satisfies the displayed difference bound. The disjoint-ball decomposition of a source above a height then gives weak $(1,1)$: its good part uses L^2, and the mean-zero bad parts use the difference bound outside the doubled balls. Interpolation gives $1 < s \le 2$. Self-adjointness supplies $s > 2$ by duality. This construction extends the same L^2 operators, rather than choosing new solutions for each s.

Apply the Dirichlet projection to the power dual of Df, using truncation and then monotone convergence when necessary. Self-adjointness and the $L^{p'}$ bound give (3.4). Finally harmonic mean-value estimates give $\|h\|_{L^p(B_{\theta r})} \le C\theta^{n/p} \|h\|_{L^p(D)}$. The projection bound controls the last norm by $C\|F\|_p$; (3.3) proves (3.5). □

The natural field Q is equivalently the coexact part $*dg$ with its relative, co-closed potential on a ball. We use its native projection characterization, retaining the whole boundary test space.

3.2 The precise div–curl and moment estimates

For scalar b, put $[b]_{\mathrm{BMO}(\mathbb{R}^n)} = \sup_B \fint_B |b - b_B|$. Only BMO functions obtained from actual Sobolev graphs are required here.

Lemma 3.2 (Local native div–curl pairing)**.** *Let $D = B_r(a)$, let $Z \in L^2(D, \mathbb{R}^k \otimes \Lambda^1)$ be interior divergence-free, and $T \in W_0^{1,2}(D, \mathbb{R}^k)$. If a bounded native $W^{1,2}$ scalar b has a whole-space BMO extension $\widehat{b}$ agreeing on D, then the native pairing agrees with its ordinary integral and*

$$\left| \int_D b\, Z : DT \right| \le C_n [\widehat{b}]_{\mathrm{BMO}(\mathbb{R}^n)} \, \|Z\|_2 \, \|DT\|_2 \,. \tag{3.6}$$

The constant is independent of the bound for b. The same estimate holds when Z is natural divergence-free and T is any native $W^{1,2}$ graph on D, with no zero-boundary condition.

Proof. Extend DT by zero. There is a solenoidal L^2 extension E of Z to $\mathbb{R}^n$ with norm at most $C_n \|Z\|_2$. To construct it, transport an extension across the sphere and correct its divergence on an annulus by solving the mean-zero weak divergence problem there; the compatibility condition is the original divergence identity. Equivalently the annular correction is defined by duality against gradients and Hilbert projection. The correction is outside D and therefore preserves the original field there.

For smooth approximations and a normalized kernel ρ_t supported in a ball of radius t, cancellation gives

$$\rho_t * (E \cdot DT)(x) = \int \nabla \rho_t(x - y) \cdot E(y)(T(y) - T_{B_{2t}(x)}) \, \mathrm{d}y.$$

Choose $s = 2n/(n+1) \in (1, 2)$. Its Sobolev conjugate is $s^* = ns/(n-s) = s/(s-1)$, so Sobolev–Poincare and Holder bound the grand maximal function of $E \cdot DT$ by

$$C\big(\mathcal{M}(|E|^s)\big)^{1/s} \big(\mathcal{M}(|DT|^s)\big)^{1/s}.$$

The maximal operator is bounded on $L^{2/s}$, so the H^1 norm is at most $C \|E\|_2 \|DT\|_2$. This is the div–curl cancellation of [3], in the precise L^2 case needed here.

Its BMO pairing can be seen directly from atoms: the level sets of the grand maximal function, decomposed into maximal disjoint balls, give a sum of mean-zero atoms, each bounded by its reciprocal ball volume, with the sum of coefficients controlled by the H^1 norm. Subtract the ball mean of $\widehat{b}$ on each atom to bound its pairing by $C[\widehat{b}]_{\mathrm{BMO}}$, and sum. Strong L^2 approximation of the weak graphs passes this estimate to the original fields. Bounded smooth truncations of the actual b show that the limit is its ordinary integral. Thus its value bound is used for identification and integrability, and does not enter the estimate. For the natural version, extend Z by zero: its full natural identity makes this extension divergence-free on all of $\mathbb{R}^n$. Extend the centered graph $T - T_D$ across the sphere and cut it off, with gradient norm at most $C \|DT\|_2$ by Poincare. The same full-space proof then applies, while the integral remains exactly the one on D. □

Lemma 3.3 (Graph BMO and value moments)**.** *The following estimates hold for fixed finite-dimensional Euclidean values.*

(i) *If $u \in W^{1,2}(B_1)$ and $\mathcal{M}_2(Du; B_1) \le \eta$, then*

$$\|u - u_{B_1}\|_{L^k(B_1)} \le C_{n,k}\eta \quad (0 < k < \infty). \tag{3.7}$$

The centered graph has scalar whole-space BMO extensions with seminorm at most $C_n\eta$ for unit scalar observables.

(ii) *For compactly supported $v \in W^{1,2}(\mathbb{R}^n)$ and $1 < p < 2$, let*

$$\Phi(x,r) = r^{p-n} \int_{B_r(x)} |Dv|^p, \qquad \Psi(t) = \sup_{x\in\mathbb{R}^n,\, 0<r\le t} \Phi(x,r). \tag{3.8}$$

On $D = B_{t/2}(a)$ every unit scalar observable of v, after a common centering, has a whole-space extension with BMO seminorm at most $C_{n,p}\Psi(8t)^{1/p}$. If $z = v_{B_{2t}(a)}$, then

$$\|v - z\|_{L^k(B_t(a))} \le C_{n,p,k}\Psi(8t)^{1/p} t^{n/k} \quad (0 < k < \infty). \tag{3.9}$$

Proof. Poincare and Holder bound the mean oscillation on each interior ball by $C\eta$. To include the boundary, extend the original weak graph evenly by inversion $x \mapsto x/|x|^2$ from the unit ball to the double ball. The derivative and Jacobian sizes are $|x|^{-2}$ and $|x|^{-2n}$; change of variables and recentering intersecting balls show that the complete truncated critical bound increases by a dimensional factor only. The two graphs have the same trace on the sphere, hence glue to a weak graph. Cut off its centered version on the double ball. Poincare controls the cutoff derivative and gives the global extension.

For clarity, the value moment argument requires only this graph BMO instance. Decompose a ball into maximal subballs where the local oscillation exceeds $A[b]_{\mathrm{BMO}}$, for fixed sufficiently large A. Their combined volume is at most half the parent volume. Off these balls differentiation bounds the oscillation by $CA[b]_{\mathrm{BMO}}$; their means differ from the parent mean by the same bound. Repeating gives $|\{|b - b_B| > jCA[b]_{\mathrm{BMO}}\} \cap B| \le 2^{-j}|B|$. Integration of $k\lambda^{k-1}$ against this distribution function proves all finite moments. Centering by u_{B_1} gives (i).

For (ii), rescale $B_{4t}(a)$. Poincare and the actual envelope bound every subball oscillation by $C\Psi(8t)^{1/p}$. Apply the same graph extension and cutoff, or use the globally defined graph with a cutoff on $B_{4t}(a)$. The distribution estimate just proved gives the moments. Changing a comparable centering mean to $v_{B_{2t}(a)}$ costs only a fixed factor. These are Sobolev-graph instances; no extension theorem for an arbitrary abstract ball-BMO function is asserted. □

4 Construction of the natural Coulomb frame

Proposition 4.1 (Critical natural frame)**.** *For fixed $n \ge 2, q$, there are $\kappa > 0, C_g < \infty$ such that every skew one-form $A \in L^2(B_1)$ with $\mathcal{M}_2(A; B_1) \le \kappa$ admits $P \in W^{1,2}(B_1, \mathrm{SO}(q))$ for which*

$$Z_i = P^T \partial_i P + P^T A_i P \tag{4.1}$$

is natural divergence-free and

$$\mathcal{M}_2(DP; B_1)^2 + \mathcal{M}_2(Z; B_1)^2 \le C_g \mathcal{M}_2(A; B_1)^2. \tag{4.2}$$

Thus Z is the literal transformed original connection, not an independently chosen field. The constants also work on every positive ball by affine transport.

For $0 \le \lambda < n$ write

$$\|F\|_{2,\lambda}^2 = \sup_{x\in B_1,\, r>0} r^{-\lambda} \int_{B_r(x)\cap B_1} |F|^2.$$

The critical exponent is $n-2$; a stronger exponent is $\nu = n-2+\gamma$, $0 < \gamma < 2$. The quantitative construction fixes $\gamma = 1$, hence $\nu = n-1$, and then takes a critical limit. Its boundary estimates are uniform for every $0 \le \lambda \le n-1$; in particular they apply separately at the original critical exponent $\lambda = n-2$.

Lemma 4.2 (Boundary estimates and the actual inverse)**.** *The ball Dirichlet, natural-gradient, and relative coexact responses are bounded on $L^{2,\lambda}$ for $0 \le \lambda < n$. If a native orthogonal frame has the natural connection (4.1) and $\mathcal{M}_2(DP) \le d_0$, then, for $\lambda = n-2, \nu$ for which the original quantities $\|A\|_{2,\lambda}$ and $\|DP\|_{2,\lambda}$ are finite,*

$$\|DP\|_{2,\lambda} + \|Z\|_{2,\lambda} \le C_{n,q,\lambda} \|A\|_{2,\lambda} + C_{n,q,\lambda} d_0 \|DP\|_{2,\lambda} . \tag{4.3}$$

Let $\mathcal{N}$ denote the native mean-zero Neumann response, so $\int D\mathcal{N}(F) : D\psi = \int F : D\psi$ for every $W^{1,2}$ test. For natural skew Z and a mean-zero native H in the stronger class,

$$\|\mathcal{N}([Z,H])\|_{W^{1,2,\nu}} \le C_{n,q,\gamma} \mathcal{M}_2(Z) \|H\|_{W^{1,2,\nu}} . \tag{4.4}$$

The norm includes the native gradient and the mean-zero value norm.

Proof. For a linear response TF, split the source at twice the radius of a ball or spherical boundary cap. The near response has energy bounded by $C \int_{B_{2r} \cap B_1} |F|^2$. The far response is harmonic there with its matching homogeneous boundary condition. Gradient comparison on the cap gives the factor $C(r/s)^n$ between its energies at scales r and s. Decomposing into annuli of radii $2^j r$ therefore bounds the Morrey norm by a geometric series with ratio $2^{-(n-\lambda)/2} < 1$. Radii comparable to or larger than one use the global energy estimate. This proves the complete truncated-ball bound, including the boundary. Poincare controls values after the mean-zero or relative zero-kernel normalization.

Here is a native construction of the nonlinear bound that retains the full boundary condition throughout. Let $\mathcal{Q}$ be the natural solenoidal projection from Lemma 3.1. Define

$$B(X,Y) = \tfrac{1}{2}(X^T Y - Y^T X), \qquad S = P^T dP = B(P, dP).$$

The last equality is the differentiated orthogonality identity and $B(X,X) = 0$. Let $Q_b = \mathcal{Q}S$ and $Q_c = \mathcal{Q}(P^T A P)$. Since Z is natural, $Z = Q_b + Q_c$. The linear response bound controls $\|Q_c\|_{2,\lambda} \le C \|A\|_{2,\lambda}$. The notation Q_b refers directly to a one-form; if a potential is desired, it is $*db$ for the unique relative co-closed $(n-2)$-form b (a zero-boundary scalar when $n = 2$).

For any bounded original critical graph P and any native graph W, let $Q_W = \mathcal{Q}(B(P, dW))$. Natural orthogonality and the natural version of Lemma 3.2 give

$$\|Q_W\|_2^2 = \int_{B_1} B(P, dW) : Q_W \le C[P]_{\mathrm{BMO}} \|DW\|_2 \|Q_W\|_2 \le C d_0 \|DW\|_2 \|Q_W\|_2 . \tag{4.5}$$

The scalar BMO coefficient here is the actual bounded graph P. The natural identity permits zero extension of Q_W, so the pairing uses every whole-ball test and has no residual normal boundary term. In particular,

$$\int_{B_1} |Q_b|^2 \le C d_0^2 \int_{B_1} |DP|^2. \tag{4.6}$$

To obtain power growth, fix $x \in B_1$, $0 < r \le 1/8$, and put $K = \|DP\|_{2,\lambda}^2 < \infty$. Even inversion extends the original graph to B_2 with both its critical bound and its r^λ energy growth multiplied by fixed constants. Choose $\chi = 1$ on $B_r(x)$, supported in $B_{2r}(x)$, with $|D\chi| \le C/r$, and let c be the extended mean on $B_{4r}(x)$. The restricted native graph $W = \chi(P - c)$ has $\|DW\|_{L^2(B_1)}^2 \le C K r^\lambda$ by Poincare on the extended ball. Alternation eliminates the quadratic cutoff term:

$$B(P - c, DW) = \chi B(P - c, DP), \qquad B(P - c, P - c) = 0.$$

Let $Q_{\mathrm{near}} = \mathcal{Q}(\chi B(P - c, DP))$. Applying (4.5) with $P - c$ and W proves $\int |Q_{\mathrm{near}}|^2 \le C d_0^2 K r^\lambda$. Subtracting c changes the full forcing only by the actual gradient $B(c, DP) = D(B(c, P))$,

which the same natural projection annihilates. Consequently $Q_{\mathrm{far}} = Q_b - Q_{\mathrm{near}}$ is the natural response to $(1-\chi)B(P-c, DP)$, a forcing that vanishes on $B_r(x) \cap B_1$.

The native gradient complement of this far response is therefore weakly harmonic there, with homogeneous Neumann boundary data on the original sphere. Its full natural identity, rather than interior divergence alone, supplies that boundary condition. The boundary harmonic gradient estimate, or the native Neumann inversion extension, gives on $B_{r/256}(x) \cap B_1$

$$|Q_{\mathrm{far}}|^2 \le Cr^{-n} \int_{B_r(x)\cap B_1} |Q_{\mathrm{far}}|^2 \quad \text{almost everywhere.}$$

Put $E_b(r) = \int_{B_r(x)\cap B_1} |Q_b|^2$. Integrate this estimate on $B_{\theta r}(x) \cap B_1$, with $0 < \theta \le 1/256$, and use the near-field energy to obtain

$$E_b(\theta r) \le C\theta^n E_b(r) + Cd_0^2 K r^\lambda. \tag{4.7}$$

Choose θ so that $C\theta^n \le \theta^\lambda/2$. At $r_0 = 1/8$, (4.6) and $\int_{B_1} |DP|^2 \le K$ give the initial bound $E_b(r_0) \le Cd_0^2 K$. Iterating (4.7), and filling radii by monotonicity, gives $E_b(r) \le Cd_0^2 K r^\lambda$ at every radius. For large radii use (4.6); recentering an intersecting ball controls any ambient center. Thus $\|Q_b\|_{2,\lambda} \le Cd_0 \|DP\|_{2,\lambda}$. Together with $Z = Q_b + Q_c$ and $DP = PZ - AP$, this proves (4.3). At the stronger exponent the assumed finite original growth is essential; no term with infinite norm is absorbed to manufacture stronger-class membership.

For (4.4), the critical coefficient is paired with an oscillation of H, rather than its absolute value. Indeed $\mathcal{N}([Z, H_D]) = 0$ for every constant matrix H_D, precisely because Z is natural. On a ball of radius r, the stronger graph has oscillation at most $C\|H\|_{W^{1,2,\nu}} r^{\gamma/2}$ by the dyadic Poincare estimate. Multiplying by the critical energy of Z gives growth $r^{n-2+\gamma}$. Near–far summation, using the same constant cancellation on each annulus, controls the mean-zero response in this stronger Morrey norm. Its value bound follows by Poincare. This proves (4.4). All graph identities are obtained by bounded smooth approximation and strong L^2 passage; the response remains the same unique native solution. □

Proof of Proposition 4.1. First suppose $A \in L^{2,\nu}$. Let Y_ν be the full native Morrey–Sobolev space of matrix-valued maps, with norm the sum of the size of the mean and $\|DH\|_{2,\nu}$. It is complete and embeds into $C^{0,\gamma/2}$: the dyadic mean estimate follows from $\int_{B_r} |DH|^2 \le Cr^{n-2+\gamma}$. The product rule and the resulting supremum bound make Y_ν a unital Banach algebra for matrix multiplication. Let X_ν be its closed real linear subspace of mean-zero skew maps. This subspace is the parameter space, not an algebra. The matrix exponential is analytic in Y_ν, agrees with the pointwise exponential, and has its actual differentiated weak graph.

Suppose a stronger-class frame is known for tA, with connection Z. For a skew mean-zero $H \in X_\nu$ and a small real s, the frame $P \exp H$ for $(t+s)A$ has connection

$$\mathcal{F}(H, s) = e^{-H} d(e^H) + e^{-H}(Z + sP^T AP)e^H.$$

The natural equation is $\mathcal{N}(\mathcal{F}(H, s)) = 0$. At $(0, 0)$ it vanishes and its derivative in H, on the skew mean-zero space, is $I + \mathcal{N}([Z, \cdot])$. By (4.4), the second term has norm less than $1/2$ if $\mathcal{M}_2(Z)$ is sufficiently small. Its inverse is the convergent Neumann series. The Banach inverse function theorem gives actual nearby stronger-class frames. Exponentiating a real skew matrix preserves special orthogonality and the literal transformed connection.

Choose d_0 so the last term of (4.3) can be absorbed at both exponents. The improved estimates are

$$\|DP\|_{2,\lambda} + \|Z\|_{2,\lambda} \le C\|tA\|_{2,\lambda} \qquad (\lambda = n-2, \nu). \tag{4.8}$$

Make the critical size of A sufficiently small that the improved critical bounds leave a strict margin below both the a priori critical budget and the Neumann-inverse threshold. Each

fieldwise local inverse-function perturbation then preserves this margin. At $t = 0$ use $P = I$. The set of $t \in [0, 1]$ with a frame and fixed critical and stronger bounds from (4.8) is relatively open by the perturbation argument and the improved bounds. It is relatively closed by weak $W^{1,2}$ compactness, strong L^2 compactness of the bounded orthogonal frames, and lower semicontinuity on every truncated ball. Strong convergence of the frames and weak convergence of their derivatives identify the connection when tested against bounded smooth fields. The natural identity then passes to all $W^{1,2}$ tests by density. The stronger growth bounds pass as well. Connectedness of $[0, 1]$ gives the frame at $t = 1$.

For critical A, extend it by zero and mollify, preserving skewness. The restrictions A_j satisfy

$$A_j \longrightarrow A \text{ strongly in } L^2(B_1), \qquad \mathcal{M}_2(A_j; B_1) \le C_n \mathcal{M}_2(A; B_1).$$

Jensen and Fubini give the bound. A convolution ball whose center falls outside the unit ball is recentered at a point of intersection, with at most twice its radius; the complete truncated definition controls it. Each mollification belongs to a stronger class, and one may fix $\gamma = 1$. Its stronger norm need not be uniform. Reduce κ to allow the dimensional convolution constant and construct frames P_j with uniform critical bounds. A joint subsequence satisfies

$$P_j \to P \text{ strongly in } L^2 \text{ and almost everywhere}, \qquad DP_j \rightharpoonup DP \text{ in } L^2.$$

The closed group $\mathrm{SO}(q)$ retains the values. Boundedness of P_j and strong convergence of A_j identify $P_j^T A_j P_j$; the strong–weak limit identifies $P_j^T DP_j$. Thus the resulting field is precisely (4.1). Its natural identity passes first for bounded smooth tests and then by density for all native tests. Lower semicontinuity on each truncated ball proves (4.2). No critical embedding into continuous functions, or approximation in the critical Morrey norm, is used. Positive affine transport preserves the critical seminorm and full natural pairing, giving every ball. □

5 The compact-target system and its actual cutoff

Work first on B_1. Fix

$$p = 1 + \frac{1}{2n}, \qquad p' = \frac{p}{p-1} = 2n + 1 > n, \qquad \delta = \frac{1}{2}. \tag{5.1}$$

Suppose the original compact-target map satisfies $-\Delta u = A \cdot Du$ and P is the constructed frame. Choose $\eta \in [0, 1]$ so that

$$\mathcal{M}_2(Du; B_1), \quad \mathcal{M}_2(DP; B_1), \quad \mathcal{M}_2(Z; B_1) \le \eta. \tag{5.2}$$

Use the actual smooth cutoff $\chi(x) = S((9/16 - |x|^2)/(17/144))$, where S is the smooth transition from zero to one. It equals one on $B_{2/3}$, vanishes outside $B_{3/4}$ and satisfies

$$0 \le \chi \le 1, \qquad |D\chi| \le B = 2^{24}, \qquad |\Delta\chi| \le L_\chi = 2^{48} + 2^{25} n. \tag{5.3}$$

These bounds follow from $|S'| \le 2^{20}$, $|S''| \le 2^{40}$ and differentiation of the displayed quadratic argument. Let

$$c = u_{B_1}, \qquad v = \chi(u - c), \qquad Dv = \chi Du + (u - c) \otimes D\chi, \tag{5.4}$$

extended by zero to $\mathbb{R}^n$. This is the original global weak graph of v. It is bounded because the target is compact.

Lemma 5.1 (The transformed equation and error)**.** *On B_1 put $F_i = P^T \partial_i v$ and $W = \sum_i Z_i F_i$. Then $F \in L^2$ and*

$$-\operatorname{div} F = W - e, \tag{5.5}$$

where the locally integrable error is the four-term expression

$$e = \sum_i (\partial_i P^T)(u-c)\partial_i \chi + 2\sum_i P^T \partial_i u \partial_i \chi + P^T(u-c)\Delta\chi + \sum_i Z_i P^T (u-c)\partial_i \chi. \tag{5.6}$$

For $0 < t \le 1/16$ and $B_t(a) \subset B_1$,

$$\|e\|_{L^1(B_t(a))} \le C\eta t^{n-1-\delta}. \tag{5.7}$$

The actual envelope (3.8) of v is finite and

$$0 \le \Psi(t) \le C\eta^p \qquad (0 < t \le 1/16). \tag{5.8}$$

Proof. Orthogonality gives $\partial_i P^T = -P^T(\partial_i P)P^T$. Test the original system with the bounded native test $P\varphi$. Product differentiation yields $-\operatorname{div}(P^T Du) = Z \cdot P^T Du$. Apply the product rule to (5.4) to obtain exactly (5.5)–(5.6). Bounded smooth graph approximation justifies this for the original weak fields; all terms on the right are L^1.

Choose $k > \max(2, n/\delta)$. The value moments and critical bounds give

$$\|u-c\|_{L^2(B_t(a))} \le C_k \eta t^{n/2-\delta}, \qquad \|Du\|_{L^2(B_t(a))} + \|DP\|_{L^2(B_t(a))} + \|Z\|_{L^2(B_t(a))} \le C\eta t^{n/2-1}.$$

The first and fourth error terms are at most $C\eta^2 t^{n-1-\delta}$, the second at most $C\eta t^{n-1}$, and the third at most $C\eta t^{n-\delta}$. Use $\eta, t \le 1$.

Holder gives $\int_{B_t} |Du|^p \le C\eta^p t^{n-p}$. For the cutoff derivative take $k > \max(p, n)$; then $t^{p-n}\int_{B_t} |u-c|^p \le C\eta^p t^{p-np/k} \le C\eta^p$. Any ball of radius at most $1/16$ meeting the support lies in $B_{7/8}$; otherwise its weak gradient is zero almost everywhere. This proves the claim for every center in $\mathbb{R}^n$, and finiteness of the supremum. □

5.1 Compensation for one and the same decomposition

Fix $0 < t \le 1/128$ and $D = B_{t/2}(a) \subset B_1$. Apply Lemma 3.1 once to $F = P^T Dv$ on D, and retain its f, w, Q, h. Write $b = \Psi(8t)^{1/p}$.

Lemma 5.2 (The same two component estimates)**.** *These fields satisfy*

$$\|Df\|_{L^p(D)} \le C\eta t^{n/p-1} b + C\eta t^{n/p-\delta}, \tag{5.9}$$

$$\|Q\|_{L^p(D)} \le C\eta t^{n/p-1} b. \tag{5.10}$$

Proof. Take $\varphi \in W_0^{1,p'}(D)$ with $\|D\varphi\|_{p'} \le 1$. Since $p' > n$,

$$\|\varphi\|_\infty \le Ct^{1-n/p'}, \qquad \|D\varphi\|_2 \le Ct^{n/2-n/p'}. \tag{5.11}$$

The same Dirichlet equation gives $\int_D Df : D\varphi = \int_D (W-e)\cdot\varphi$. Move the derivative of v in W onto the matrix test constructed from $P\varphi$. In coordinates its derivative has the two terms $DP\,\varphi + P\,D\varphi$. The resulting pairings are scalar observables of v against Z and this derivative. Centering constants vanish by the divergence identity and zero trace. The product test is native and zero-boundary since both P and φ are bounded. Lemmas 3.2 and 3.3 give

$$\left|\int_D W\cdot\varphi\right| \le Cb\,\|Z\|_2\,(\|DP\|_2\,\|\varphi\|_\infty + \|D\varphi\|_2) \le C\eta t^{n/p-1} b.$$

The actual bounded cutoff identifies this pairing with the ordinary integral. The error is bounded by $C\eta t^{n-1-\delta}t^{1-n/p'} = C\eta t^{n/p-\delta}$. Dirichlet duality proves (5.9).

For Q, let $H \in L^{p'}(D)$ be a one-form of norm at most one and T its natural solenoidal projection. Then $\|T\|_{p'} \le C$. Self-adjointness and the full natural identity give, with $z = v_{B_{2t}(a)}$,

$$\int_D Q : H = \int_D P^T Dv : T = -\int_D (v - z) \cdot \sum_i \partial_i P\, T_i. \tag{5.12}$$

For smooth bounded approximations this is integration by parts; the native product test is justified by boundedness of P and v. The constant term vanishes because the natural identity can test the frame itself. For general duals use approximation and boundedness of the same natural projection. Let $s = 2p/(2-p)$; then $1/2 + 1/p' + 1/s = 1$. Holder and (3.9) bound the last integral by

$$C \|DP\|_2 \|T\|_{p'} \|v - z\|_s \le C\eta t^{n/2-1} b t^{n/s} = C\eta t^{n/p-1} b.$$

All limiting products are integrable by this same three-factor bound. Duality proves (5.10) for the original selected Q; no field is reselected for a dual test. □

Lemma 5.3 (One-ball decay)**.** *For $0 < \theta \le 1/2$,*

$$\int_{B_{\theta t/2}(a)} |Dv|^p \le C\left((\eta^p + \theta^n)\Psi(8t)t^{n-p} + \eta^p t^{n-p\delta}\right). \tag{5.13}$$

Proof. Orthogonality gives $|F| = |Dv|$ almost everywhere. Insert the same component bounds into (3.5). Use $(a+b)^p \le 2^p(a^p + b^p)$ and $\|F\|^p_{L^p(D)} \le \Psi(8t)t^{n-p}$, with $n - p \ge 0$. This proves the displayed inequality directly from the original data. □

6 All-center iteration and the representative

Lemma 6.1 (Envelope improvement)**.** *There are $\eta_* > 0, C$ such that $\eta \le \eta_*$ implies*

$$\Psi(t) \le C\eta^p t^\beta \quad (0 < t \le 1/16), \qquad \beta = p(1-\delta) = p/2. \tag{6.1}$$

For every $a \in \mathbb{R}^n$ and $r > 0$,

$$\int_{B_r(a)} |Dv|^p \le C\eta^p r^{n-p+\beta} = C\eta^p r^{n-p/2}. \tag{6.2}$$

Proof. To pass from one ball to all centers, fix $r \le \theta t/2$ and set $t' = 2r/\theta \le t$. If $B_r(a)$ misses the support of Dv, its integral is zero. Otherwise a is within distance r of $B_{3/4}$; since $t \le 1/128$, the closed ball of radius $t'/2$ at a is contained in B_1. Apply (5.13) at t', multiply by $r^{p-n} = (\theta t'/2)^{p-n}$, and use monotonicity of Ψ and $(t')^\beta \le t^\beta$. Taking the supremum over all such a, r gives

$$\Psi(\theta t/2) \le D(\eta^p \theta^{p-n} + \theta^p)\Psi(8t) + D\eta^p \theta^{p-n} t^\beta \quad (0 < t \le 1/128). \tag{6.3}$$

Every supremum is finite by (5.8); neither attainment nor continuity is needed. The constants in this argument can be chosen without any real root, logarithm, or search. Let $D > 0$ be the rational coefficient in (6.3), let M_0 be the integer initial bound for $\Psi(1/16)/\eta^p$, and let J

be the integer global bound $r^{p-n}\int_{B_r(a)}|Dv|^p \le J\eta^p$. These specific coefficients are given in Section 7 and Appendix A. Put

$$H = 512D, \qquad \vartheta = (2+8H)^{-2}, \qquad \eta_* = \frac{\vartheta^n}{4(1+H)}, \qquad \sigma = \vartheta/16, \tag{6.4}$$

$$A_0 = D/\vartheta^n, \qquad B_0 = 16M_0, \qquad C_s = \frac{B_0 + 2A_0/\sigma}{\sigma}, \qquad C_{\mathrm{pow}} = 1 + C_s + 16J. \tag{6.5}$$

Here $0 < \vartheta \le 1/2$, $0 < \eta_* \le 1$, and $0 < \beta = p/2 < 1$. The identity $\sqrt{\vartheta} = (2+8H)^{-1}$ gives, for $p \ge 1$,

$$H\vartheta^p \le \tfrac{1}{2}\vartheta^{p/2}, \qquad H\eta^p\vartheta^{p-n} \le \tfrac{1}{2}\vartheta^{p/2} \quad (0 \le \eta \le \eta_*).$$

For the second inequality use $\eta^p \le \eta_*$ and $\vartheta^p \le \vartheta^{p/2}$. Since $16^\beta \le 16$ and $2\cdot 16^\beta D \le H$, substituting $t = s/8$ in (6.3) gives

$$\Psi(\sigma s) \le \tfrac{1}{2}\sigma^\beta \Psi(s) + A_0\eta^p s^\beta \quad (0 < s \le 1/16). \tag{6.6}$$

Indeed $D\vartheta^{p-n} \le D\vartheta^{-n} = A_0$. Also $(1/16)^\beta \ge 1/16$, so the initial bound is $\Psi(1/16) \le B_0\eta^p(1/16)^\beta$. Iteration at $s_j = \sigma^j/16$ and monotonicity between successive radii give the coefficient

$$\frac{B_0 + A_0/(\sigma^\beta - \sigma^\beta/2)}{\sigma^\beta} \le \frac{B_0 + 2A_0/\sigma}{\sigma} = C_s,$$

where $\sigma^\beta \ge \sigma$ was used twice. Thus (6.1) holds with C_{pow}. Multiplying by r^{n-p} proves (6.2) for $r \le 1/16$. For $r \ge 1/16$, the original global bound and $r^\beta \ge 1/16$ imply $J \le 16Jr^\beta \le C_{\mathrm{pow}}r^\beta$. This gives the same coefficient for every positive radius. □

Lemma 6.2 (Direct mean-limit representative)**.** *Suppose $v \in W^{1,2}(\mathbb{R}^n, \mathbb{R}^q)$, $1 \le p \le 2$, $0 < \alpha < 1$, and $\int_{B_r(a)}|Dv|^p \le K^p r^{n-p+p\alpha}$ for all $a, r > 0$. Then v has a continuous representative with $[\widetilde{v}]_{C^{0,\alpha}} \le C_{n,p,\alpha}K$.*

Proof. Poincare and Holder give $⨍_{B_r(a)}|v - v_{B_r(a)}| \le CKr^\alpha$. Successive means at radii $r, r/2, r/4, \ldots$ differ by at most $CK(r/2^j)^\alpha$. Their limits define $\widetilde{v}(a)$ at every point, with tail at most CKr^α. Lebesgue differentiation identifies the limit with v almost everywhere. For distinct a, b, take $r = 2|a-b|$ and compare their radius-r means through a ball of comparable radius containing both. Poincare bounds the mean difference and the two tails by CKr^α. This proves the pointwise Hölder estimate and continuity everywhere. □

Proposition 6.3 (The compact-target representative)**.** *The original system and its constructed frame satisfying (5.2) with $\eta \le \eta_*$ have a unique continuous representative on $B_{2/3}$, with Hölder exponent $1/2$ and seminorm at most $C\eta$. On a ball $B_\rho(a)$ the corresponding conclusion is on $B_{2\rho/3}(a)$ with bound $C\eta\rho^{-1/2}$.*

Proof. Apply (6.2) and the preceding mean-limit lemma. On $B_{2/3}$ the cutoff is one, so $c + \widetilde{v}$ represents the original u. It lies in N at every point: continuity and closedness of N would otherwise give an open ball outside N, contradicting almost-everywhere target membership. Two continuous representatives agree everywhere because a nonzero difference persists on an open set of positive measure. Under $x = a + \rho y$, the weak gradient has the factor ρ, the critical quantities are invariant, and the Hölder seminorm has the factor $\rho^{-1/2}$. □

7 A complete ledger of rational constants

The estimates above admit the following particular witnesses. They are conservative admissible bounds; no optimality is asserted. Every coefficient entering the effective threshold and Hölder bound is specified by finite natural-number or rational arithmetic. Appendix A expands the entire recursion, including its Hardy, projection, boundary and moment coefficients, into a self-contained algorithm. Its function names agree with the numerical definitions in the formal development.

7.1 The geometric certificate and the effective theorem

The geometric input consists of an actual function $G : \mathbb{R}^q \to \mathcal{L}(\mathbb{R}^q, \mathbb{R}^q)$ and a rational $\ell \geq 0$ such that

$$G \in C^1_c, \qquad G(y) = Q(y) \ (y \in N), \qquad \|DG(y)\| \leq \ell \quad (y \in \mathbb{R}^q). \tag{7.1}$$

The last inequality covers every ambient point, including the extension and cutoff region. The extension theorem and compactness prove existence of such data for every compact embedded C^2 target; an integer larger than the finite derivative bound is a rational choice of ℓ. This is an existence statement about an abstract embedded target. To evaluate numerical bounds for a specified embedding one supplies and certifies (7.1). The present theorem does not supply an algorithm extracting this certificate from an arbitrary abstract manifold structure. After certification, the analytic constants require only n, q, ℓ and no computation of u or of its Coulomb frame.

Theorem 7.1 (Rational quantitative witnesses)**.** *Under the hypotheses of Theorem 1.2, fix any certificate (7.1). Define κ, C_g, θ, C_r by the ledger below and ε_*, C by (8.1). The conclusions of Theorem 1.2 hold with exactly these values and exponent $1/2$. They are the functions*

`effectiveStationaryEnergyThreshold`
`effectiveStationaryHolderConstant`

evaluated at (n, q, ℓ) in Appendix A. If the target is smooth, the same values also give the Hölder estimate for the smooth representative.

The proof is the quantitative tracking below followed by the assembly in Section 8. This effective interface exposes certified geometric data. The existential target-only theorem obtains those data inside its proof, so its original hypotheses are unchanged.

7.2 Elementary and moment coefficients

Write $V_n = |B_1(0) \subset \mathbb{R}^n|$. This auxiliary real volume is eliminated from every numerical coefficient by the following integer bounds. The coordinate cube of side $1/(n+1)$ centered at zero lies in B_1, and $B_1 \subset [-1, 1]^n$. Consequently

$$(n+1)^{-n} \leq V_n \leq 2^n, \qquad V_n^{-s} \leq 1 + (n+1)^n \quad (0 \leq s \leq 1). \tag{7.2}$$

Coordinate Sobolev integration gives coefficient n at exponent $n/(n-1)$; the norm-power argument gives $np(n-1)/(n-p)$ for $1 \leq p < n$. At the CLMS lower exponent $2n/(n+1)$ this coefficient is exactly $2n$. All coordinate norm comparisons are bounded by n, so no norm of an unspecified basis occurs. The scalar transition bounds and radial cutoff bounds are those in (5.3); the scaled unit-to-two cutoff has gradient bound 2^{22}.

For $k \in \mathbb{N}$ put

$$T_k = 1 + 24\, 2^k k!.$$

This bounds the geometric tail moment for every $0 < \varrho \le 1/2$:

$$1 + \sum_{m \ge 0} (m+2)^k \varrho^m \le T_k.$$

Indeed $x^k/k! \le e^x$ at $x = (m+2)/2$, $e < 3$ and $e^{1/2} \le 7/4$ bound each summand by $3\,2^k k!(7/8)^m$, whose sum is $24\,2^k k!$. Apply this to the actual maximal-function distribution tails, of ratios $1/4$ or $2^{-p} \le 1/2$, after the dyadic mean estimates. For example, with $D_n = \texttt{effectivePointwiseConstantn}$ the integer moment coefficient is exactly

$$I_{n,k} = T_k D_n^k (2^n + 5^n). \tag{7.3}$$

The lower-p counterpart uses `criticalLpPointwiseTailMajorantn` in place of D_n. The coexact component needs only its fourth moment: for our fixed p, $2p/(2-p) \le 4$. The error term uses $k = 2n+3$ and ordinary Holder reduction. Compact-target boundedness ensures admissibility of the ordinary pairings and does not enter these coefficients.

Ball-average telescoping at every $1/2 \le \alpha \le 1$ has the uniform coefficient $512\,2^n$. Combining this with (7.2) and the actual Sobolev Poincare coefficient gives

$$S_n = 1 + 512\,2^n\big(2^{n+1}(n+1)(1+(n+1)^n)\big). \tag{7.4}$$

It controls the local Morrey representative at $\alpha = 1/2$ and the global representative at every exponent in $[1/2, 1]$ used by the supercritical native tests. In particular, at $r = 2n+1$ their essential value bound has integer coefficient

$$U_n = 4S_n(1 + 2^n).$$

This follows from the exterior zero anchor, $4^\alpha \le 4$, and $V_n^{1/2-1/r} \le 1 + 2^n$.

7.3 Actual pairing and projection coefficients

The local div–curl coefficient is

$$H_n = 2\,D_{\mathrm{at}}(n) K_{\mathrm{gr}}(n),$$

where $D_{\mathrm{at}} = \texttt{rationalHardyPairingConstant}$ and $K_{\mathrm{gr}} = \texttt{divcurlGrandKernelMajorant}$ are expanded in the appendix. The latter comes from the explicit coordinate Sobolev coefficient, scale normalization and maximal-function estimate. The former comes from the actual Whitney partition, normalized mass, compensation and atom coefficients. The mass normalization uses $V_n^{-1} \le (n+1)^n$. Thus this is a bound for the actual ordinary pairing in Lemma 3.2, rather than an assumed Hardy representation coefficient. Native boundary extension and its cutoff multiply it by the explicit integer extension and mean-oscillation factors displayed in the appendix.

For clarity, the numerical good-lambda-to-moment step has the following literal form. Given the actual off-support coefficient c, set

$$\begin{aligned} A(n,c) &= 1 + 64^n + 16c64^n, & D(n) &= 1 + 4\,5^n 64^n 1536^n, \\ d(N,A,D) &= 4(1+D)(1+A)^N, \\ T(n,N,A,D) &= 2(1+A)^N d(N,A,D)^N 2^N\, N\, 5^n. \end{aligned}$$

Choosing the good-lambda parameter $1/d$ proves the actual density moment bound with coefficient T for every $2 \le s \le N$. The contraction factor is at most one half. Here $s = (2n+1)/2$ and $N = n+1$ lie in this window. The Neumann coefficient is

$$c_N = \max\{64^n, (1 + 2\,2^{2n} + 2(n-2)^2 2^{2(n-2)} 4^n) 16^n\}.$$

The Dirichlet coefficient is

$$c_D = 64^n + \big(1 + 2^{2n}[2 + 8n(n+2)^2(1+4^n)2^{n+1}]\big)16^n.$$

These are obtained from the genuine Kelvin boundary gradient estimates, including the natural boundary tests. Put $T_N = T(n, n+1, A(n, c_N), D(n))$ and $T_D = T(n, n+1, A(n, c_D), D(n))$. Self-adjoint duality acts on the same fixed projections. It supplies the coexact dual coefficient $2^{2n+1}(1 + T_N)$, the Dirichlet gradient moment coefficient T_D, and the same Dirichlet test's essential value coefficient $U_n(1 + T_D)$. All weak identities and ordinary pairing integrabilities refer to those same tests.

7.4 The critical gauge constants

The following rational recurrence function records the full-boundary power estimate. For nonnegative f, d, e, let

$$t_f = \frac{1}{256(1+4f)}, \qquad b_e = e8^{n-1},$$
$$P_n(f,d,e) = 2^{n-1}\left(1 + b_e + \frac{b_e + d(2+2f)/(t_f^{n-1} - 2ft_f^n)}{t_f^{n-1}}\right).$$

Its denominators are positive since $2ft_f \le 1/2$. The comparison $t_f^{n-1} \le t_f^\lambda$ for $0 \le \lambda \le n-1$ makes the same number valid at all those exponents, including $n-2$. The natural and alternating near/far coefficients in its arguments are the specific integers and rationals listed in the appendix.

Let $F_n = \texttt{effectiveNativeNeumannPowerConstantn}$, $L_n = 2(1+F_n)$, $A_n = 2(4L_n + 2)$ and $J_n = 2A_n + 2$. Set $C_2 = 2J_n$ and

$$C_g = 4C_2\, 2^{n-2}. \tag{7.5}$$

Here A_n is the absorbed gradient coefficient, J_n the joint frame/connection coefficient, and C_2 their common two-exponent majorant. Denote the alternating logarithmic-response coefficient by $L_{\log}(n, q)$ and the stronger mean-zero natural response norm bound by $L_{\mathrm{nat}}(n)$; their exact definitions use P_n and are expanded in the appendix. Define

$$a = \frac{1}{8(1 + L_{\log}(n,q))}, \qquad j = \frac{1}{2L_{\mathrm{nat}}(n)(1 + 2(q+1))},$$
$$k_s = \min\left\{\frac{j}{1+C_2}, \frac{a/4}{1+C_2}\right\}, \qquad \kappa = k_s/2^n. \tag{7.6}$$

The first threshold gives $4L_{\log}a^2 \le 1/2$ in the finite original growth absorption. The second makes the actual linearized natural Neumann operator have norm at most $1/2$, with inverse norm at most 2. The scalar-path threshold ensures both these inequalities throughout continuation. The local inverse-function neighborhood may depend on the field; it is used only for openness of the closed scalar path. It does not enter k_s. Finally the skew mollifications inflate the critical amplitude by $\sqrt{2^{n-2}} \le 2^n$, so κ works uniformly before the critical limit. Adding the two limiting critical squares gives exactly (7.5).

7.5 Cutoff, Hodge and representative constants

Use $B = 2^{24}$ and $L = L_\chi = 2^{48} + 2^{25}n$. For ease of reading the following symbols name fully expanded appendix functions; the arguments n, q, B, L are displayed when needed:

$$\begin{aligned} C_f &= 1 + F_{\mathrm{sys}}(n,q)(1+T_D) + U_n(1+T_D), \\ C_{f,\chi} &= 1 + C_f + C_f E_{1/2}(n,B,L), \\ C_Q &= 1 + (1 + 2^{2n+1}(1+T_N))(1 + I_{\mathrm{cap},4}(n)), \\ C_H &= 1 + C_{f,\chi} + C_Q, \\ C_{\mathrm{one}} &= 1 + (16 + 256\,2^n)(1 + 5(1+C_H)^2), \\ D &= 2^n C_{\mathrm{one}}. \end{aligned}$$

Here F_{sys} is the matrix native forcing coefficient, defined by `effectiveSystemForcingConstant`. The four-term error coefficient $E_{1/2}$ is `effectiveHalfCutoffErrorConstant`; the actual cutoff fourth-moment coefficient $I_{\mathrm{cap},4}$ is `criticalLpCappedFourMomentMajorant`. Each formula is given literally in the appendix. These formulas bound the same Dirichlet and coexact fields in the same Hodge decomposition, and then its harmonic remainder. The bound $x^{1/p} \le 1 + x$ for $x \ge 0$, $p \ge 1$ removes the nonrational roots from the coefficients; $x^p \le (1+x)^2$ for $1 \le p \le 2$ removes variable exponents in the component combination.

Let K_χ be the coefficient `effectiveCutoffCriticalConstant` at $(n, 1, B)$ and put

$$J = (1 + 2^n)(1 + K_\chi), \qquad M_0 = 1 + J + 2^{n+1}(n+1)(1 + (n+1)^n).$$

Insert these and the displayed D into (6.4)–(6.5). The given-frame smallness and representative coefficient are exactly

$$\theta = \eta_* = \frac{\vartheta^n}{4(1 + 512D)}, \qquad C_r = S_n(1 + C_{\mathrm{pow}}). \tag{7.7}$$

They are positive rationals and $\theta \le 1$. The radius-two local representative with coefficient S_n recovers B_1 for the cutoff graph; its plateau recovers the original map on $B_{2/3}$. The target bound, frame and local IFT witnesses have disappeared from the numerical values. The only target-specific parameter remaining in (8.1) is the certified ℓ, with q recording the ambient dimension.

Remark 7.2 (An arithmetic example). For the numerical inputs $n = 2$, $q = 3$, $\ell = 1$, the exact functions give

$$\varepsilon_* \approx 2.227969608538 \times 10^{-6428}, \qquad C \approx 2.951399297746 \times 10^{5772}.$$

Exact integer cross multiplication verifies $10^{-6428} \le \varepsilon_* < 10^{-6427}$ and $10^{5772} \le C < 10^{5773}$. This is a function evaluation at specified inputs, not a certification that any particular embedded target admits (7.1) with $\ell = 1$. The size of these bounds reflects repeated conservative majorants; it has no optimality meaning.

8 Proof of the small-energy theorem

Proof of Theorem 1.2. Fix a certified G, ℓ as above and the rational constants κ, C_g, θ, C_r of Section 7. Put

$$M = (1 + 4q\ell^2)4^{n-2}, \qquad K = (1 + C_g)(1 + M), \qquad \varepsilon_* = (\min(\kappa, \theta)/K)^2, \qquad C = 2C_r K. \tag{8.1}$$

All quantities are positive rational numbers. Write $e = \Theta_u(a, R)$, $\rho = 3R/4$, $\zeta = \sqrt{M}\sqrt{e}$ and $\eta = (1 + C_g)\zeta$. Since $\sqrt{M} \le 1 + M$ and $C_g \ge 0$, $e \le \varepsilon_*$ implies

$$0 \le \zeta \le \kappa, \qquad 0 \le \eta \le \theta \le 1, \qquad \eta \le K\sqrt{e}.$$

Lemma 2.3 bounds the joint original gradient and connection critical squares by $Me = \zeta^2$ on $B_\rho(a)$. Proposition 4.1 constructs a native natural frame on this same ball, with the joint DP, Z square bounded by $C_g\zeta^2$. Both the original gradient square and these two frame squares are bounded by η^2, because $C_g \le (1 + C_g)^2$.

Proposition 6.3, with its precise rational witnesses θ, C_r, applies to the actual original map and this frame. The resulting ball is exactly $B_{2\rho/3}(a) = B_{R/2}(a)$, and

$$[\widetilde{u}]_{C^{0,1/2}(B_{R/2}(a))} \le C_r\eta(3R/4)^{-1/2} \le 2C_rK\sqrt{e}\,R^{-1/2}.$$

We used $(3/4)^{-1/2} \le 2$, so the last coefficient is precisely C in (8.1). Target membership and uniqueness come from the same proposition. The argument includes $e = 0$; the resulting representative is constant and the right side is zero. The chosen G, ℓ depend only on the embedded target, and all analytic coefficients depend only on n, q. Thus all constants precede U, u, Du, a, R, as required. The certified pair exists for every target in Theorem 1.2; no geometric certificate is an additional premise of that existential theorem. □

9 The smooth-target corollary

Corollary 9.1. *If N is a compact smooth embedded target, the representative of Theorem 1.2 is smooth on the same half ball. It retains (1.5), all-point target membership, and uniqueness among continuous representatives. No quantitative bound for all higher derivatives is asserted.*

Proof. We use the standard interior smooth bootstrap for locally Hölder weak solutions of a smooth quadratic-gradient Poisson equation: if $w \in W^{1,2}_{\mathrm{loc}}(V)$ is locally Hölder, A is a smooth bilinear coefficient depending on w, and

$$\Delta w = \sum_i A(w)(\partial_i w, \partial_i w)$$

in the actual weak sense, then w is smooth locally. In this criterion the initial weak gradient and weak equation are retained. Analytically, one freezes the coefficient on a sufficiently small ball, uses the small oscillation of w and harmonic comparison to obtain the first derivative regularity, and then applies interior elliptic estimates and Schauder bootstrap to the smooth quadratic coefficient. The statement is a regularity theorem about the weak equation, and is not the false claim that an arbitrary continuous quadratic system can be differentiated classically without this first regularity step. We use this proved bootstrap rather than reproving general interior elliptic theory here; its complete formal implementation is the theorem `HolderToCK.holderQuadraticPoisson_smooth` in `HolderToCK/Frozen/HolderQuadraticPoissonSmooth.lean`.

For the present map, choose a smooth ambient normal-projection extension G. Lemma 2.1 gives $\Delta u = -\sum_i DG(u)[\partial_i u]\partial_i u$. Almost-everywhere replacement by the Hölder representative preserves the original weak gradient and the equation on the half ball; the coefficient becomes $A(\widetilde{u}) = -DG(\widetilde{u})$, which is smooth in its argument. All hypotheses of the stated bootstrap are therefore supplied internally by Theorem 1.2, the original variation condition, and smooth target geometry. The conclusion is smoothness of this same representative on every interior ball of the half ball. Continuous uniqueness identifies the local smooth versions. The original quantitative Hölder estimate is unchanged. □

10 Correspondence with the formal proof

The public theorem is `StationaryHarmonicMaps.stationary_small_energy_regularity`. It chooses ε, α, C after the fixed dimension and target, and before the open domain, original map and weak gradient, center, and positive radius. The proof chooses $\alpha = 1/2$. Its conclusion covers the entire half ball, all-point target membership, the original energy square-root factor, and uniqueness among all continuous almost-everywhere representatives.

The weak-harmonicity premise is the original tangent-variation condition. The proof internally constructs the normal-projection system, complete Morrey bounds, and native frame, then supplies every premise of the compensated representative result. No Hodge component, compensation bound, recurrence, or regular representative is assumed. The following paths are under `StationaryHarmonicMaps/`, except for explicitly prefixed `HolderToCK/` paths relative to the repository root. Intermediate analytic lemmas may have inputs produced by preceding theorems; all such inputs are supplied inside the main proof.

Mathematical step	Formal source
Embedded target and original variations	`Foundation/C2Target.lean`; `Foundation/WeakHarmonicity.lean`
Projection geometry and original equation	`Frozen/NormalProjectionCharacterization.lean`; `Frozen/VariationalWeakPoissonEquivalence.lean`
Monotonicity and complete Morrey bound	`Frozen/Monotonicity.lean`; `Frozen/StationaryMorreyBound.lean`; `Providers/SmallEnergyRegularity.lean`
Ball projections and all powers	`Analysis/NeumannGradientProjection.lean`; `Analysis/NeumannOperatorAllLp.lean`; `Analysis/HodgeLowPComparison.lean`
Actual graph BMO and moments	`Analysis/BoundaryCriticalBMOExtension.lean`; `Analysis/CriticalLpMorreyRealMoments.lean`; `Analysis/CappedEnvelopeValueMoments.lean`
Solenoidal extension and native pairing	`Analysis/SolenoidalBallExtension.lean`; `Analysis/LocalDivCurlHardy.lean`; `Analysis/LocalDivCurlContinuousPairing.lean`
Boundary Morrey estimates and absorption	`Analysis/CoulombBoundaryAbsorption.lean`; `Analysis/CoulombTwoExponentAbsorption.lean`
Actual stronger-class perturbation	`Analysis/NativeCoulombPerturbations.lean`; `Analysis/StrongerNativeCoulombExistence.lean`
Joint critical limit and native frame	`Analysis/CriticalNativeCoulombExistence.lean`; `Analysis/CriticalNativeCoulombBallExistence.lean`
Same cutoff and compensated fields	`Analysis/SystemCutoffCompensation.lean`; `Analysis/SystemCoexactCompensation.lean`; `Analysis/SystemCompensatedHodgeFields.lean`
One-ball decay and capped envelope	`Analysis/OneBallCompensatedDecay.lean`; `Analysis/CappedEnvelopeImprovement.lean`; `Analysis/CompensatedCutoffPowerGrowth.lean`
Mean-limit representative and scaling	`Analysis/LocalLpMorreyRepresentative.lean`; `Analysis/CompensatedSystemRepresentative.lean`
Original theorem assembly	`Providers/SmallEnergyRegularity.lean`; `Frozen/SmallEnergyRegularity.lean`
Certified geometric data	`Geometry/EffectiveNormalProjectionData.lean`
Rational threshold and coefficient	`Analysis/EffectiveStationaryConstants.lean`; `Analysis/EffectiveEnergyAmplitudes.lean`
Effective stationary theorem and smooth corollary	`Frozen/EffectiveSmallEnergyRegularity.lean`; `Frozen/EffectiveSmallEnergySmoothRegularity.lean`; `Providers/EffectiveSmallEnergyRegularity.lean`; `EffectiveRegularity.lean`

Mathematical step	Formal source
Smooth-target corollary	`Analysis/SmoothHolderRepresentative.lean`; `Providers/SmallEnergySmoothRegularity.lean`; `Frozen/SmallEnergySmoothRegularity.lean`

Lean's kernel checks the proof. Its axiom closure contains only `propext`, `Classical.choice`, and `Quot.sound`. There is no custom analytic axiom, `sorryAx`, or native trust bypass. Imported library theorems and the pinned formal dependency are themselves formal proofs, rather than unformalized analytic assumptions. The repository README records the reproducible build and exact theorem verification.

The effective exports, under `StationaryHarmonicMaps`, are

```
stationary_small_energy_regularity_effective
stationary_small_energy_smooth_regularity_effective.
```

Their certified geometric inputs are exactly (7.1); their threshold and Hölder coefficient are the two literal rational functions in Appendix A. The existential geometric-data theorem supplies such a pair for every target in the original C^2 theorem.

The scope is this closed small-energy theorem and its actual formal dependencies. It does not certify a separate arbitrary ball-BMO extension theorem, an arbitrary unbounded-system theorem, or a $W^{1,p}_{\mathrm{loc}}$ representative theorem on all possible $p \geq 1$. The actual Sobolev-graph and $W^{1,2}$ instances proved here are the ones consumed by the main theorem. Citations identify the method, and do not replace missing formal premises.

A Complete exact-arithmetic evaluator

The following self-contained Lean 4 program defines all arithmetic used by Theorem 7.1: 114 definitions, of which 59 return natural numbers and 55 return rational numbers. Dependencies precede their users. It imports only the basic rational and factorial libraries, and no stationary-map, gauge, or analytic estimate from this project. Namespace prefixes are omitted inside this single namespace; the function bodies are the literal formulas used by the certified coefficients above.

Read `Nat` as $\mathbb{N}$, `Rat` as $\mathbb{Q}$, `a^k` as an integer power, `k.factorial` as $k!$, and `Inv.inva` as a^{-1}. Natural-number subtraction is truncated at zero. The theorem uses $n \geq 2$, so $n-1, n-2$ in its formulas have ordinary nonnegative values. Rational operations are exact; the proof above establishes positivity of the denominators on the prescribed parameter range. No real root, transcendental evaluation, unspecified norm, or noncomputable choice occurs in these numerical functions. The runtime imports enable evaluation of the same exact integer and rational operations. The evaluator output is not used as a premise of an analytic proof.

```
module

public import Mathlib.Data.Rat.Cast.Order
public import Mathlib.Data.Nat.Factorial.Basic
meta import Mathlib.Data.Rat.Cast.Order
meta import Mathlib.Data.Nat.Factorial.Basic

@[expose] public section
namespace SmallEnergyConstants

def criticalBound (n : Nat) (strongerBound : Rat) : Rat :=
  2 * strongerBound * 2 ^ (n - 2)

def naturalPowerScale (farBound : Rat) : Rat :=
```

```
  1 / (256 * (1 + 4 * farBound))

def neumannBoundaryBound (n : Nat) : Nat :=
  1 + 2 * 2 ^ (2 * n) + 2 * (n - 2) ^ 2 * 2 ^ (2 * (n - 2)) * 4 ^ n

def effectiveNeumannOffSupportConstant (n : Nat) : Nat :=
  max (64 ^ n) (neumannBoundaryBound n * 16 ^ n)

def effectiveNativeNeumannPowerConstant (n : Nat) : Rat :=
  let C : Rat := effectiveNeumannOffSupportConstant n
  let theta := naturalPowerScale (2 * C)
  1 + (1 + (2 + 4 * C) / (theta ^ (n - 1) - 4 * C * theta ^ n)) / theta ^ (n - 1)

def effectiveCoexactPowerConstant (n : Nat) : Rat :=
  2 * (1 + effectiveNativeNeumannPowerConstant n)

def effectiveCoulombAbsorptionConstant (n : Nat) : Rat :=
  2 * (4 * effectiveCoexactPowerConstant n + 2)

def effectiveCoulombJointConstant (n : Nat) : Rat :=
  2 * effectiveCoulombAbsorptionConstant n + 2

def effectiveCoulombCriticalConstant (n : Nat) : Rat :=
  2 * effectiveCoulombJointConstant n

def effectiveCoulombTwoExponentConstant (n : Nat) : Rat :=
  effectiveCoulombCriticalConstant n

def effectiveStrongerCoulombGradientConstant (n : Nat) : Rat :=
  2 * effectiveCoulombTwoExponentConstant n

def effectiveCoulombFrameGradientConstant (n _q : Nat) : Rat :=
  criticalBound n (effectiveStrongerCoulombGradientConstant n)

def criticalAmplitude (n : Nat) (strongerAmplitude : Rat) : Rat :=
  strongerAmplitude / 2 ^ n

def scalarPathAmplitude (absorption inverse bound : Rat) : Rat :=
  min (inverse / (1 + bound)) ((absorption / 4) / (1 + bound))

def absorptionAmplitude (boundaryBound : Rat) : Rat :=
  1 / (8 * (1 + boundaryBound))

def naturalPowerBound (n : Nat) (farBound nearBound energyBound : Rat) : Rat :=
  let theta := naturalPowerScale farBound
  let beta := energyBound * 8 ^ (n - 1)
  2 ^ (n - 1) * (1 + beta +
    (beta + nearBound * (2 + 2 * farBound) /
      (theta ^ (n - 1) - 2 * farBound * theta ^ n)) / theta ^ (n - 1))

def effectiveAlternatingFarConstant (n : Nat) : Rat :=
  2 * (effectiveNeumannOffSupportConstant n : Rat)

def effectiveBoundaryLocalizedTestConstant (n : Nat) : Nat :=
  2 + 2 * n * (2 ^ 22) ^ 2 * (2 ^ (n + 1) * 16)

def alternatingNearBound (n : Nat) (compensatedBound : Rat) : Rat :=
  compensatedBound * (effectiveBoundaryLocalizedTestConstant n : Rat) *
    (2 ^ (n - 1) * (1 + 2 ^ (2 * n))) * 4 ^ (n - 1)

def effectiveBoundaryReflectionFactor (n : Nat) : Nat :=
  2^(n-2)*(1+2^(2*n))

def effectiveGlobalCriticalMeanOscillationConstant (n : Nat) : Nat :=
  1 + 2 ^ (n + 1) * (n + 1) * (1 + (n + 1) ^ n)

def effectiveDyadicMeanConstant (n : Nat) : Nat :=
  1 + 2 ^ n * effectiveGlobalCriticalMeanOscillationConstant n +
    16 ^ n * effectiveGlobalCriticalMeanOscillationConstant n

def effectivePointwiseConstant (n : Nat) : Nat :=
  effectiveDyadicMeanConstant n + 2 ^ (2 * (n + 1)) * (n + 1) + 1
```

```
def geometricMomentMajorant (k : Nat) : Nat := 1 + 24 * 2^k * k.factorial

def effectiveIntegerMomentConstant (n k : Nat) : Nat :=
  geometricMomentMajorant k * effectivePointwiseConstant n ^ k * (2 ^ n + 5 ^ n)

def effectiveLocalValueConstant (n : Nat) : Nat :=
  (1 + 2 ^ n) * (1 + effectiveIntegerMomentConstant n (2 * n + 1))

def effectiveAllRadiusValueConstant (n : Nat) : Nat :=
  effectiveLocalValueConstant n + effectiveIntegerMomentConstant n 2

def effectiveCutoffCriticalConstant (n A B : Nat) : Nat :=
  1 + 2 * A ^ 2 * 2 ^ (n - 2) + 2 * n * B ^ 2 * effectiveAllRadiusValueConstant n

def effectiveCutoffGlobalMeanConstant (n A B : Nat) : Nat :=
  effectiveGlobalCriticalMeanOscillationConstant n * (1 + effectiveCutoffCriticalConstant n A B)

def effectiveBoundaryCriticalMeanConstant (n : Nat) : Nat :=
  effectiveCutoffGlobalMeanConstant n 1 (2^23) *
    (1+effectiveBoundaryReflectionFactor n)

def divcurlLocalSobolevMajorant (n : Nat) : Nat :=
  (1+2*n)*(n+2+4*n*2^22*(n+1)*2^n)

def divcurlLowerKernelMajorant (n : Nat) : Nat :=
  1+divcurlLocalSobolevMajorant n*(n+1)

def divcurlScaledKernelMajorant (n : Nat) : Nat :=
  divcurlLowerKernelMajorant n*2^n*(1+4^n)

def divcurlDyadicKernelMajorant (n : Nat) : Nat :=
  divcurlScaledKernelMajorant n*2^n*2^n

def divcurlGrandKernelMajorant (n : Nat) : Nat :=
  divcurlDyadicKernelMajorant n*(4*(n+1)*5^n)

def rationalHardyPairingConstant (n : Nat) : Rat :=
  let B : Rat := ((1 + 3 * (60 : Rat)^n) * (2 : Rat)^22) / 5
  let mass : Rat -> Rat := fun b => 2*b*(40 : Rat)^(n+1)*(60 : Rat)^n*((n : Rat)+1)^n
  let D := mass B
  let A := max ((2 : Rat)^22*(10 : Rat)^(n+1)*((n : Rat)+1)^n) D
  2 * ((4*D + 2*mass (4*B) + 2*A) * (2*(70 : Rat)^n))

def effectiveLocalDivcurlPairingConstant (n : Nat) : Rat :=
  2*rationalHardyPairingConstant n*(divcurlGrandKernelMajorant n:Rat)

def effectiveNativeBoundaryTestConstant (n : Nat) : Nat :=
  (2 + 2 * n * (2 ^ 22) ^ 2 * (2 ^ (n + 1) * 4)) * (1 + 2 ^ (2 * n))

def effectiveBoundaryScalarConstant (n : Nat) : Rat :=
  effectiveLocalDivcurlPairingConstant n * (effectiveBoundaryCriticalMeanConstant n : Rat) *
    (1+(effectiveNativeBoundaryTestConstant n : Rat))

def effectiveBoundaryBilinearConstant (n m : Nat) (b : Rat) : Rat :=
  effectiveBoundaryScalarConstant n * (1+(m : Rat)*b)

def effectiveCenteredCoexactConstant (n m : Nat) (coefficientBound : Rat) : Rat :=
  effectiveBoundaryBilinearConstant n m coefficientBound ^ 2

def effectiveAlternatingNearConstant (n m : Nat) (coefficientBound : Rat) : Rat :=
  alternatingNearBound n (effectiveCenteredCoexactConstant n m coefficientBound)

def effectiveAlternatingPowerConstant (n m : Nat) (coefficientBound : Rat) : Rat :=
  naturalPowerBound n (effectiveAlternatingFarConstant n)
    (effectiveAlternatingNearConstant n m coefficientBound)
    (effectiveCenteredCoexactConstant n m coefficientBound)

def effectiveCoulombLogarithmicPowerConstant (n q : Nat) : Rat :=
  effectiveAlternatingPowerConstant n (q*q) ((q + 1 : Nat) : Rat)

def effectiveCoulombAbsorptionAmplitude (n q : Nat) : Rat :=
```

```
  absorptionAmplitude (effectiveCoulombLogarithmicPowerConstant n q)

def inverseAmplitude (operatorBound bilinearBound : Rat) : Rat :=
  1 / (2 * operatorBound * (1 + bilinearBound))

def meanZeroBound (n : Nat) : Nat := 1 + 2 ^ (n + 1)

def effectiveNaturalFarConstant (n : Nat) : Nat :=
  2 * effectiveNeumannOffSupportConstant n

def effectiveBoundaryStrongerConstant (n : Nat) : Nat :=
  1 + (512 * 2 ^ n) * effectiveGlobalCriticalMeanOscillationConstant n *
    (1 + 2 ^ (n - 1) * (1 + 2 ^ (2 * n)))

def effectiveStrongerMeanConstant (n : Nat) : Nat :=
  2 * effectiveBoundaryStrongerConstant n

def effectiveNaturalInitialEnergyConstant (n : Nat) : Nat :=
  effectiveStrongerMeanConstant n ^ 2

def effectiveNaturalNearEnergyConstant (n : Nat) : Nat :=
  effectiveBoundaryStrongerConstant n ^ 2

def effectiveNaturalNeumannPowerConstant (n : Nat) : Rat :=
  naturalPowerBound n (effectiveNaturalFarConstant n : Rat)
    (effectiveNaturalNearEnergyConstant n : Rat) (effectiveNaturalInitialEnergyConstant n : Rat)

def effectiveNaturalNeumannOperatorConstant (n : Nat) : Rat :=
  (meanZeroBound n : Rat) * (1 + effectiveNaturalNeumannPowerConstant n)

def effectiveNaturalNeumannInverseAmplitude (n : Nat) (bilinearBound : Rat) : Rat :=
  inverseAmplitude (effectiveNaturalNeumannOperatorConstant n) bilinearBound

def effectiveSkewNeumannAmplitude (n q : Nat) : Rat :=
  effectiveNaturalNeumannInverseAmplitude n (2 * ((q + 1 : Nat) : Rat))

def effectiveScalarCoulombSmallness (n q : Nat) : Rat :=
  scalarPathAmplitude (effectiveCoulombAbsorptionAmplitude n q)
    (effectiveSkewNeumannAmplitude n q) (effectiveCoulombTwoExponentConstant n)

def effectiveCoulombFrameSmallness (n q : Nat) : Rat :=
  criticalAmplitude n (effectiveScalarCoulombSmallness n q)

def effectiveCompensationCutoffGradientBound : Nat := 2^24

def effectiveCompensationCutoffLaplacianBound (n : Nat) : Nat := 2^48+2^25*n

def effectiveHodgeLowPowerConstant (n : Nat) : Nat :=
  16+256*2^n

def criticalLpMeanOscillationMajorant (n : Nat) : Nat :=
  2^(n+1) * (n+1) * (1 + (n+1)^n)

def criticalLpDyadicMeanMajorant (n : Nat) : Nat :=
  1 + (2^n + 16^n) * criticalLpMeanOscillationMajorant n

def criticalLpPointwiseTailMajorant (n : Nat) : Nat :=
  criticalLpDyadicMeanMajorant n + 2^(n+1) * 2^(n+1) * (n+1) + 1

def criticalLpIntegerMomentMajorant (n k : Nat) : Nat :=
  geometricMomentMajorant k * criticalLpPointwiseTailMajorant n^k * (2^n + 5^n)

def criticalLpFourMomentMajorant (n : Nat) : Nat :=
  (1 + criticalLpIntegerMomentMajorant n 4) * (1 + 2^n)

def criticalLpCappedFourMomentMajorant (n : Nat) : Nat :=
  criticalLpFourMomentMajorant n * 2^n

def densityGoodLambdaTailFactor (n : Nat) : Nat :=
  1 + 4*5^n*64^n*1536^n

def densityGoodLambdaAmplitude (n c : Nat) : Nat :=
```

```
  1 + 64^n + 16*c*64^n

def effectiveNeumannGoodLambdaAmplitude (n : Nat) : Nat :=
  densityGoodLambdaAmplitude n (effectiveNeumannOffSupportConstant n)

def uniformGoodLambdaDenominator (N A D : Nat) : Nat :=
  4*(1+D)*(1+A)^N

def uniformGoodLambdaMomentMajorant (n N A D : Nat) : Nat :=
  2*(1+A)^N*(uniformGoodLambdaDenominator N A D)^N*2^N*N*5^n

def effectiveNeumannDimensionalMomentConstant (n : Nat) : Nat :=
  uniformGoodLambdaMomentMajorant n (n+1) (effectiveNeumannGoodLambdaAmplitude n)
    (densityGoodLambdaTailFactor n)

def effectiveCoexactDimensionalDualConstant (n : Nat) : Nat :=
  2^(2*n+1)*(1+effectiveNeumannDimensionalMomentConstant n)

def effectiveCoexactCompensatedConstant (n : Nat) : Nat :=
  1+effectiveCoexactDimensionalDualConstant n

def effectiveSystemCoexactConstant (n : Nat) : Nat :=
  1+effectiveCoexactCompensatedConstant n*(1+criticalLpCappedFourMomentMajorant n)

def effectiveHalfLocalValueConstant (n : Nat) : Nat :=
  (1+2^n)*(1+effectiveIntegerMomentConstant n (2*n+3))

def effectiveHalfCutoffErrorConstant (n B K : Nat) : Nat :=
  1+B*(n+1)*(2*(1+effectiveHalfLocalValueConstant n)+2*(1+2^n))+
    K*((1+2^n)*(1+effectiveHalfLocalValueConstant n))

def oddKelvinBoundaryEnergyMajorant (n : Nat) : Nat :=
  1 + 2^(2*n) * (2 + (8*n*(n+2)^2) * ((1+4^n)*2^(n+1)))

def sameDirichletOffSupportMajorant (n : Nat) : Nat :=
  64^n + oddKelvinBoundaryEnergyMajorant n * 16^n

def dirichletGoodLambdaAmplitude (n : Nat) : Nat :=
  densityGoodLambdaAmplitude n (sameDirichletOffSupportMajorant n)

def dirichletFixedHighMomentMajorant (n : Nat) : Nat :=
  uniformGoodLambdaMomentMajorant n (n+1)
    (dirichletGoodLambdaAmplitude n) (densityGoodLambdaTailFactor n)

def explicitLpMorreyRepresentativeMajorant (n : Nat) : Nat :=
  1 + 512 * 2 ^ n * (2 ^ (n + 1) * (n + 1) * (1 + (n + 1) ^ n))

def nativeZeroBoundaryFixedSupercriticalMajorant (n : Nat) : Nat :=
  4 * explicitLpMorreyRepresentativeMajorant n * (1 + 2^n)

def boundedDirichletFixedTestMajorant (n : Nat) : Nat :=
  nativeZeroBoundaryFixedSupercriticalMajorant n * (1+dirichletFixedHighMomentMajorant n)

def effectiveLocalMeanOscillationCutoffConstant (n : Nat) : Nat :=
  2*(32^n+1+2^23*(2*2^n+2*4^n))

def effectiveLpMeanOscillationConstant (n : Nat) : Nat :=
  2^(n+1)*(n+1)*(1+(n+1)^n)

def effectiveLocalLpCriticalCutoffConstant (n : Nat) : Nat :=
  1+effectiveLocalMeanOscillationCutoffConstant n*effectiveLpMeanOscillationConstant n

def effectiveCompactTargetEnvelopePairingConstant (n : Nat) : Rat :=
  effectiveLocalDivcurlPairingConstant n*(effectiveLocalLpCriticalCutoffConstant n:Rat)

def effectiveSystemMatrixSupercriticalConstant (n q : Nat) : Nat :=
  1+2*(q+1)*(nativeZeroBoundaryFixedSupercriticalMajorant n+1+2^n)

def effectiveSystemForcingConstant (n q : Nat) : Rat :=
  2*effectiveCompactTargetEnvelopePairingConstant n*
    (effectiveSystemMatrixSupercriticalConstant n q:Rat)*(1+q)
```

```
def effectiveSystemDirichletConstant (n q : Nat) : Rat :=
  1+effectiveSystemForcingConstant n q*(1+(dirichletFixedHighMomentMajorant n : Rat))+
    (boundedDirichletFixedTestMajorant n : Rat)

def effectiveSystemCutoffDirichletConstant (n q B L : Nat) : Rat :=
  1+effectiveSystemDirichletConstant n q+
    effectiveSystemDirichletConstant n q*(effectiveHalfCutoffErrorConstant n B L : Rat)

def effectiveSystemCompensatedHodgeConstant (n q B L : Nat) : Rat :=
  1+effectiveSystemCutoffDirichletConstant n q B L+(effectiveSystemCoexactConstant n : Rat)

def effectiveOneBallCompensatedConstant (n q B L : Nat) : Rat :=
  1+(effectiveHodgeLowPowerConstant n : Rat)*
    (1+5*(1+effectiveSystemCompensatedHodgeConstant n q B L)^2)

def effectiveCappedEnvelopeImprovementConstant (n q B L : Nat) : Rat :=
  effectiveOneBallCompensatedConstant n q B L*2^n

def effectiveCutoffPowerAmplitude (n q B L : Nat) : Rat :=
  512*effectiveCappedEnvelopeImprovementConstant n q B L

def rationalEnvelopeScale (G : Rat) : Rat := Inv.inv (2 + 8 * G) ^ 2

def rationalEnvelopeSmallness (n : Nat) (G : Rat) : Rat :=
  rationalEnvelopeScale G ^ n / (4 * (1 + G))

def effectiveCutoffPowerSmallness (n q B L : Nat) : Rat :=
  rationalEnvelopeSmallness n (effectiveCutoffPowerAmplitude n q B L)

def effectiveGivenFrameSmallness (n q : Nat) : Rat :=
  effectiveCutoffPowerSmallness n q effectiveCompensationCutoffGradientBound
    (effectiveCompensationCutoffLaplacianBound n)

def rationalMorreyFactor (n q : Nat) (L : Rat) : Rat :=
  (1+(q:Rat)*(2*L)^2)*(4:Rat)^(n-2)

def rationalAmplitudeMajorant (n q : Nat) (L G : Rat) : Rat :=
  (1+G)*(1+rationalMorreyFactor n q L)

def rationalEnergyThreshold (n q : Nat) (L G kappa theta : Rat) : Rat :=
  (min kappa theta/rationalAmplitudeMajorant n q L G)^2

def effectiveStationaryEnergyThreshold (n q : Nat) (L : Rat) : Rat :=
  rationalEnergyThreshold n q L (effectiveCoulombFrameGradientConstant n q)
    (effectiveCoulombFrameSmallness n q) (effectiveGivenFrameSmallness n q)

def effectiveCutoffLpEnvelopeConstant (n A B : Nat) : Nat :=
  (1+2^n)*(1+effectiveCutoffCriticalConstant n A B)

def effectiveCutoffCappedEnvelopeConstant (n A B : Nat) : Nat :=
  1+effectiveCutoffLpEnvelopeConstant n A B+effectiveLpMeanOscillationConstant n

def effectiveCutoffPowerScale (n q B L : Nat) : Rat :=
  rationalEnvelopeScale (effectiveCutoffPowerAmplitude n q B L)

def rationalRecurrenceMajorant (sigma A B : Rat) : Rat := (B + 2 * A / sigma) / sigma

def effectiveCutoffPowerGrowthConstant (n q A B L : Nat) : Rat :=
  1+rationalRecurrenceMajorant (effectiveCutoffPowerScale n q B L/16)
    (effectiveCappedEnvelopeImprovementConstant n q B L/effectiveCutoffPowerScale n q B L^n)
    (16*(effectiveCutoffCappedEnvelopeConstant n A B : Rat))+
      16*(effectiveCutoffLpEnvelopeConstant n A B : Rat)

def effectiveCutoffHolderConstant (n q A B L : Nat) : Rat :=
  (explicitLpMorreyRepresentativeMajorant n : Rat)*(1+effectiveCutoffPowerGrowthConstant n q A B L)

def effectiveGivenFrameHolderConstant (n q : Nat) : Rat :=
  effectiveCutoffHolderConstant n q 1 effectiveCompensationCutoffGradientBound
    (effectiveCompensationCutoffLaplacianBound n)

def rationalHolderCoefficient (n q : Nat) (L G H : Rat) : Rat :=
  2*H*rationalAmplitudeMajorant n q L G
```

```
def effectiveStationaryHolderConstant (n q : Nat) (L : Rat) : Rat :=
  rationalHolderCoefficient n q L (effectiveCoulombFrameGradientConstant n q)
    (effectiveGivenFrameHolderConstant n q)

end SmallEnergyConstants
```

For example, append the following commands to evaluate the two exact rational values in Remark 7.2:

```
open SmallEnergyConstants
#eval effectiveStationaryEnergyThreshold 2 3 1
#eval effectiveStationaryHolderConstant 2 3 1
```

The first fraction has a 10-digit numerator and a 6438-digit denominator; the second has a 5778-digit numerator and denominator 67453. The values coincide exactly with the public constant functions, including all four integers of these two fractions.

References

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